UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT 101: ALGEBRA AND TRIGONOMETRY CODE 1 QUIZ I
ATTEMPT ALL QUESTIONS. TIME ALLOWED: 30 MINS

NAME:..... INDEX NUMBER:.....

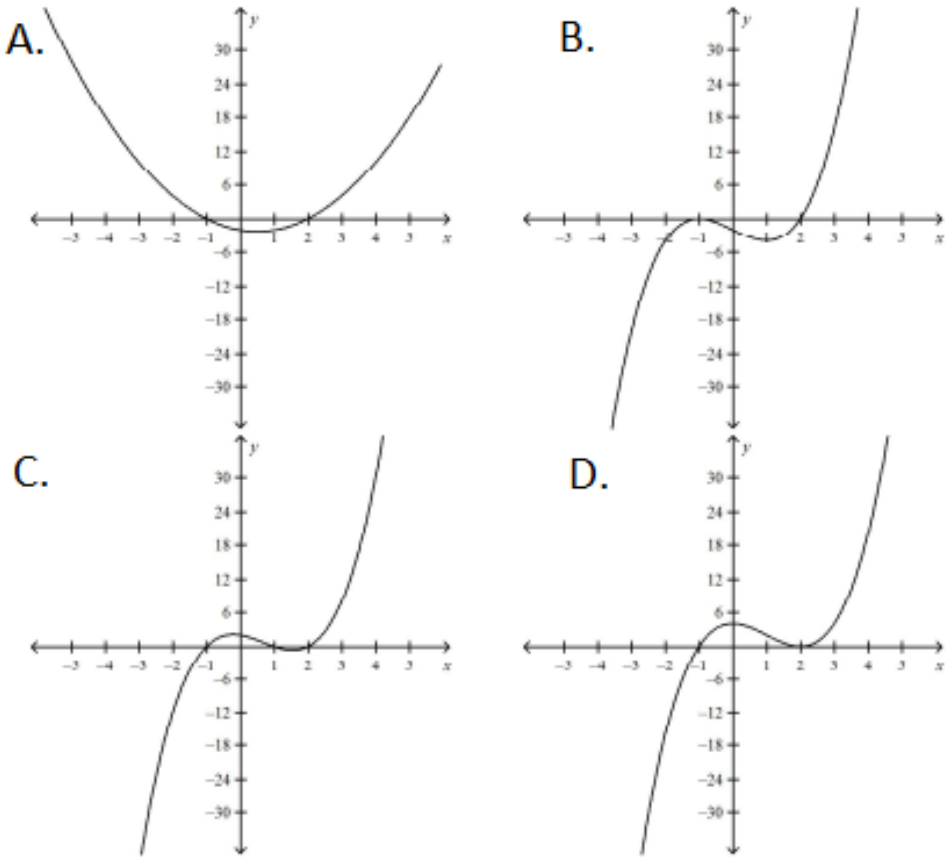
1. Find the value of x for which the function $f(x) = 2 - 3x - 4x^2$ has the greatest value.

   A. $\frac{41}{16}$ B. $-\frac{41}{16}$ C. $\frac{3}{8}$ D. $-\frac{3}{8}$

2. If $(x+2)$ and $(2x-1)$ are factors of ax^2+3x+b. Determine the relation between a and b.

   A. $a > b$ B. $a < b$ C. $a = b$ D. $a \leq b$

3. Which of the following graphs is a possible sketch of the function $g(x) = (x+1)^2(x-2)$.



4. If $M \cup N = M$, find $M \cap N$.

A. M B. N C. ϕ D. M'

5. Given that m and n are the roots of the quadratic equation $(5+\sqrt{2})x^2 - (4+\sqrt{5})x + (8+2\sqrt{5}) = 0$.
Find the value of $\frac{2mn}{(m+n)}$.

A. 2 B. 4 C. 7 D. 8

6. If α and β are the roots of the equation $x^2 - q(1+x) - r = 0$. Determine the value of $(1+\alpha)(1+\beta)$.

A. $1-r$ B. $1+r$
C. $q-r$ D. $q+r$

7. Let k_1 and k_2 be the graphs of the functions $p(x)$ and $q(x)$ respectively, with gradients m_1 and m_2. Pick the odd one out.

A. k_1 and k_2 are perpendicular if and only if $m_1 = m_2$.
B. k_1 and k_2 are parallel if and only if $m_1 m_2 = -1$.
C. k_1 and k_2 are perpendicular if and only if $\frac{1}{m_1} = -m_2$.
D. k_1 and k_2 are parallel if and only if $\frac{m_1}{m_2} = -1$.

8. Given that $U = \{1, 3, 4\}$ and $V = \{x : x \in \mathbb{R} \text{ and } x^2 - 7x + 12 = 0\}$.
Find the relationship between U and V.

A. $U = V$ B. $U \equiv V$
C. $U \subset V$ D. $V \subset U$

9. The sum of any two irrational numbers .....

A. is always a rational number
B. is always an irrational number
C. may be a rational or irrational number
D. is always an integer

10. Find the solution set for $0 < -x/2 < 3$.

A. $(-6, 6)$
B. $(0, 6)$
C. $(-6, 0)$
D. $(-\infty, -6)$

11. If $S = \{1, 4\},\ R = \{2, 3\}\ ,\ T = \{3, 5\}$, find $(S \times R) \cap (S \times T)$.

A. $\{(1, 3),\ (4, 3)\}$
B. $\{(1, 3),\ (1, 5),\ (2, 5)\}$
C. $\{(1, 3),\ (2, 5)\}$
D. None of the above

12. Given that $(x+2)$ is a factor of the polynomial $f(x) = x^3+4x^2+7x+k$, where k is a constant. Determine the value of k.

A. 0
B. 2
C. -6
D. 6

13. Which of the following numbers is the product of a non-zero rational number and an irrational number?

A. Irrational number
B. Whole number
C. Rational number
D. Natural number

14. Which of the following functions define a one-to-one mapping in the set of real numbers?

A. A quadratic function
B. A constant linear function
C. A cubic function
D. A non-constant linear function

15. Find the turning point of the parabola $p(x) = x^2 + x - 5$.

A. $(-\frac{1}{2}, -\frac{21}{4})$
B. $(-\frac{1}{2}, \frac{19}{4})$
C. $(\frac{1}{2}, \frac{19}{4})$
D. $(-\frac{1}{2}, \frac{21}{4})$

16. Find the zeros of the polynomial $y = 3(x+4)(x+1)(x+3)$?

A. $-4, -1, 3$
B. $4, 1, -3$
C. $3, -3, 4, 1$
D. $-4, -1, -3$

17. Given that $f(x) \equiv ax^3 - x^2 - 5x + b$, where a and b are constants. When $f(x)$ is divided by $(x-2)$, the remainder is 36 and when divided by $(x+2)$ the remainder is 40. Find the values of a and b.

A. $a = 1,\ b = 42$
B. $a = 42,\ b = 1$
C. $a = -1,\ b = 42$
D. $a = -42,\ b = -1$

18. Given that $p, q, r \in \mathbb{R}$, state the trichotomy law.

A. $p = q$, $p \geq q$ or $p \leq q$
B. $p = q$, $p > q$ or $p < q$
C. if $p > q,\ q > r,$ then $p > r$
D. if $p \geq q,\ q \geq r,$ then $p \geq r$

19. Solve the inequality $|2x - 3| \leq 4$.

A. $[-\frac{1}{2}, \frac{7}{2}]$
B. $(-\frac{1}{2}, \frac{7}{2}]$
C. $[-\frac{1}{2}, \frac{7}{2})$
D. $(-\frac{1}{2}, \frac{7}{2})$

20. Solve the inequality $\left|\frac{3-x}{2}\right| \geq 3$

A. $(-\infty, -3]$ or $[9, \infty)$
B. $(-\infty, -3)$ or $(9, \infty)$
C. $(-\infty, -3]$ or $(9, \infty)$
D. $(-\infty, -3)$ or $[9, \infty)$